

# Sean W. Evans

Sean@SeanWEvans.com | +1 (516) 330-2517 | GitHub | Website | LinkedIn

## Summary

---

Systems engineer focused on HPC, database internals, and ML infrastructure. Built GPU simulation engines, PostgreSQL extensions, OCR pipelines, and CI/CD security tooling in C++, CUDA, Rust, Python, and SQL.

## Experience

---

### Data Conversion Laboratory

Remote

*Machine Learning Engineer*

*May 2021 – Feb 2025*

- Designed a Detectron2-based document segmentation pipeline that reduced manual classification by 50%.
- Built a CNN-RNN LaTeX OCR system for high-accuracy equation and document recognition.
- Automated document styling with NLP and the Microsoft Office SDK, reducing a 40-hour workflow to 2 hours.
- Implemented an OpenCV-based checkbox detector achieving greater than 96% accuracy across thousands of formats.
- Developed high-throughput processing pipelines capable of handling millions of pages per week.

### Data Conversion Laboratory

Remote

*Software Engineer*

*Sep 2020 – May 2021*

- Architected a PDF cleaning and OCR preprocessing system scaling to more than 100,000 pages per week per server.
- Automated LaTeX and JATS XML correction workflows, replacing a full-time manual process.
- Developed DOCX-to-XML conversion tooling using Microsoft Office Interop and C#.

### Data Conversion Laboratory

Queens, NY

*Lead Technology Analyst*

*Sep 2019 – Sep 2020*

- Led workflow optimization efforts, mentored junior engineers, and coordinated technical requirements with stakeholders.

### Data Conversion Laboratory

Queens, NY

*Technology Analyst*

*May 2018 – Sep 2019*

- Developed custom data-conversion tools for production support and maintained high-volume enterprise pipelines.

## Selected Projects

---

- **Lockstep** – Data-oriented systems programming language for high-throughput, deterministic compute pipelines, enforcing a straight-line SIMD execution model and static memory topology for maximum CPU vectorization.
- **fluid-sims** – Collection of high-performance, CUDA-accelerated fluid dynamics and physics simulators, including SPH, hypersonic flow, and reaction-diffusion systems.
- **pg\_gpt2** – Complete implementation of the GPT-2 architecture entirely inside PostgreSQL, featuring a native tensor engine, autograd, and AdamW optimization for end-to-end training and inference via SQL.
- **WarpDB** – GPU-accelerated SQL query engine that leverages CUDA JIT compilation to dynamically generate optimized kernels for high-performance analytical workloads and multi-GPU data processing.
- **Baloo** – Collection of essential UNIX utilities written in pure x86\_64 assembly using direct syscalls. No libc, no dependencies.

## Skills

---

**Languages:** Rust, C++, C, NASM, CUDA, Python, SQL, TypeScript

**Systems:** Linux, multiprocessing, GPU kernels, distributed systems, consensus algorithms, compiler tooling, AST analysis

**Databases:** PostgreSQL internals, extension development, vector search, ETL pipelines, XML processing

**ML / CV:** PyTorch, Detectron2, OpenCV, OCR pipelines

## Education

---

**Long Island University**

January 2026

B.S. in Mathematics